Exam 1 - AGST 5014

Name:

2023-03-10

1. The ANOVA from a factorial randomized complete block experiment output is shown below. In this experiment, we evaluated 100 cultivars, 4 locations, and 3 replications.
a) Fill in the blanks.

Source	DF	SS	MS	F	P
Block	2	30	15	15.79	0.0000001
cultivar	99	138	1.39	1.46	0.0038
location	3	90	30	31.58	0.0
interaction	297	300	1.01	1.06	0.26
Error	798	754	0.95		
Total	1199	1312			

b) Interpret the anova results above.

Block was significantly different, which indicates that the our experimental units are not homogeneous and blocking was important. The interaction is not significant (at an alpha of 0.05), so we can interpret each main effect independently. Both main effects (cultivar and location) are statistically significant, so we reject the null hypothesis that all the cultivars have the same effect. the same for location, we reject the null hypothesis that the four locations have the same effect, i.e., at least one location has different mean compared to the other locations.

2. What are the steps required for running experiments (what would be your pipeline for running an experiment from beginning to end)? There is no clear cut on the number of steps taken.

- Formulate the hypothesis (research question)
- Define the variables of interest (response, factors)
- Define the number of experimental units and reps
- Choose the experimental design
- Design (randomize) the experiment
- Conduct exploratory data analysis
- Analyze the experiment
- Check model assumptions
- Run means comparison
- Interpret the results

3. Interpret the following results:

a) Shapiro-Wilk's test

The null hypothesis is that the data (we should look at the residuals) follow a normal distribution. Thus, we reject the null hypothesis of normality.

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals
## W = 0.95202, p-value < 2.2e-16
```

b) Levene's test

The null hypothesis is that the residuals are homogeneously distributed. Thus, we do not reject the null hypothesis. Regarding variance of the residuals, we are safe to proceed with ANOVA.

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  9  0.8863 0.5536
##       20
```

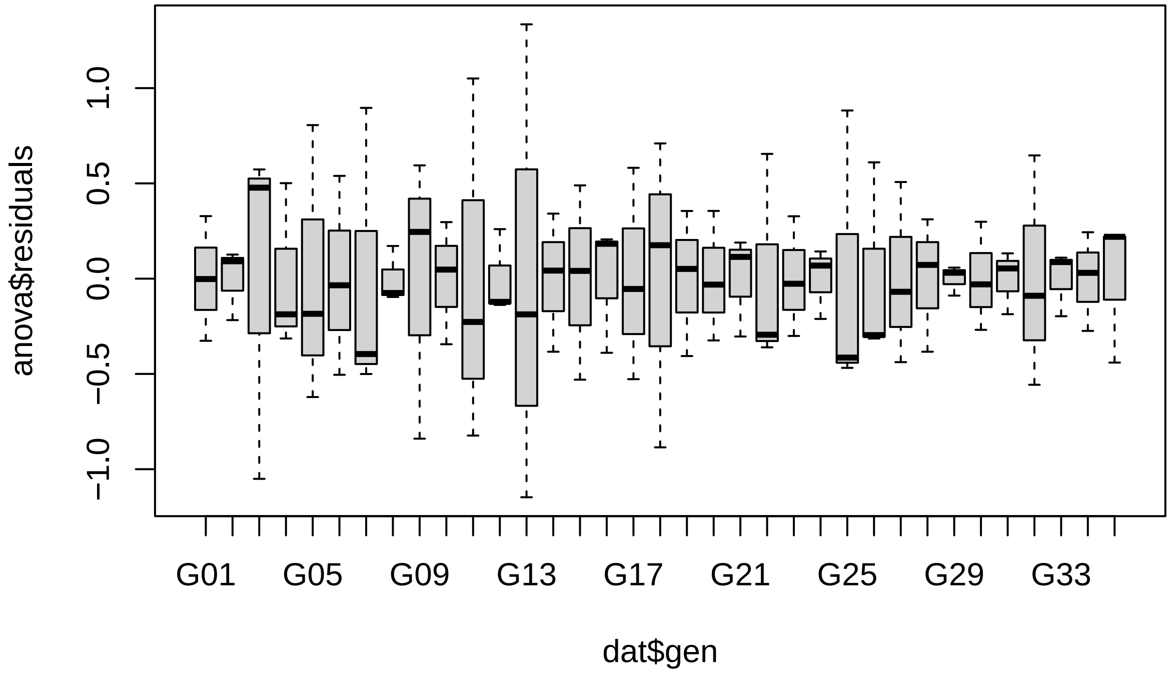
c) Tukey 1 degree of freedom (additivity test)

The null hypothesis is that the right model is additive. Thus, we do not reject the null hypothesis of additivity. I.e., the interaction is not important and we can use a model without the interaction effect.

```
##
## Tukey's one df test for additivity
## F = 5.1626778   Denom df = 1    p-value = 0.2639426
```

4. Can we proceed with the analysis below? Do you see anything wrong? If yes, what would you propose?

```r
anova <- aov(yield ~ rep + gen, dat)
boxplot(anova$residuals ~ dat$gen)
```




```r
dat$residuals <- anova$residuals
car::leveneTest(residuals ~ gen, data=dat)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group 34  0.5532 0.9701
##       70
```

The boxplot of the residuals per treatment (gen) indicate that the variances are not homogeneous, i.e., some levels have a much larger variance than others. Therefore, we could try to transform the data or use another approach such as weighted least squares, or even mixed-models (for clarification, you didn't have to write mixed models here since it is a topic of the next exam). However, in this particular data set, there are only 3 replicates, and the boxplot is not a good measurement for the variation. The *Brown–Forsythe* test indicated that there was no heterogeneity of variances.

5. Interpret the following results:

```r
q5_aov <- aov(Y ~ treat, data = ex1q5)
summary(q5_aov)
```

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## treat        5 1133.4  226.68   30.85 3.16e-08 ***
## Residuals   18  132.2    7.35
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
tukey(ex1q5$Y, ex1q5$treat, 18 , 132.2, alpha = 0.05)
```

```
##
## Tukey's test
```

```
## ------------------------------------------------------------------------
## Groups Treatments Means
## a      0.37    82.75
## ab     0.51    77
##  b     0.71    75
##  b     1.02    71.75
##   c    1.4     65
##   c    1.99    62.75
## ------------------------------------------------------------------------
```

There is a significant effect of treatment, which indicate at least one level of treatment is statistically different from the others. The Tukey's test indicate that the treatment 0.37

6. Below there is the analysis of an experiment that measured the effect of different herbicides on different crops. This was a CRD where all the treatments had the same number of replications. Based on the ANOVA table for the sums of squares type III, fill in the table for the sums of squares type I

ANOVA Table (Type III tests)

Source	DF	SS	MS	F	P
crop	1	455	-	59.32	0.0001
herbicide	3	1333	-	173.74	0
interaction	3	29	-	3.39	0.056
Error	66	507			

ANOVA Table (Type I tests)

Source	DF	SS	MS	F	P
crop	1	455	-	59.32	0.0001
herbicide	3	1333	-	173.74	0
interaction	3	29	-	3.39	0.056
Error	66	507			

7. Answer the following questions:

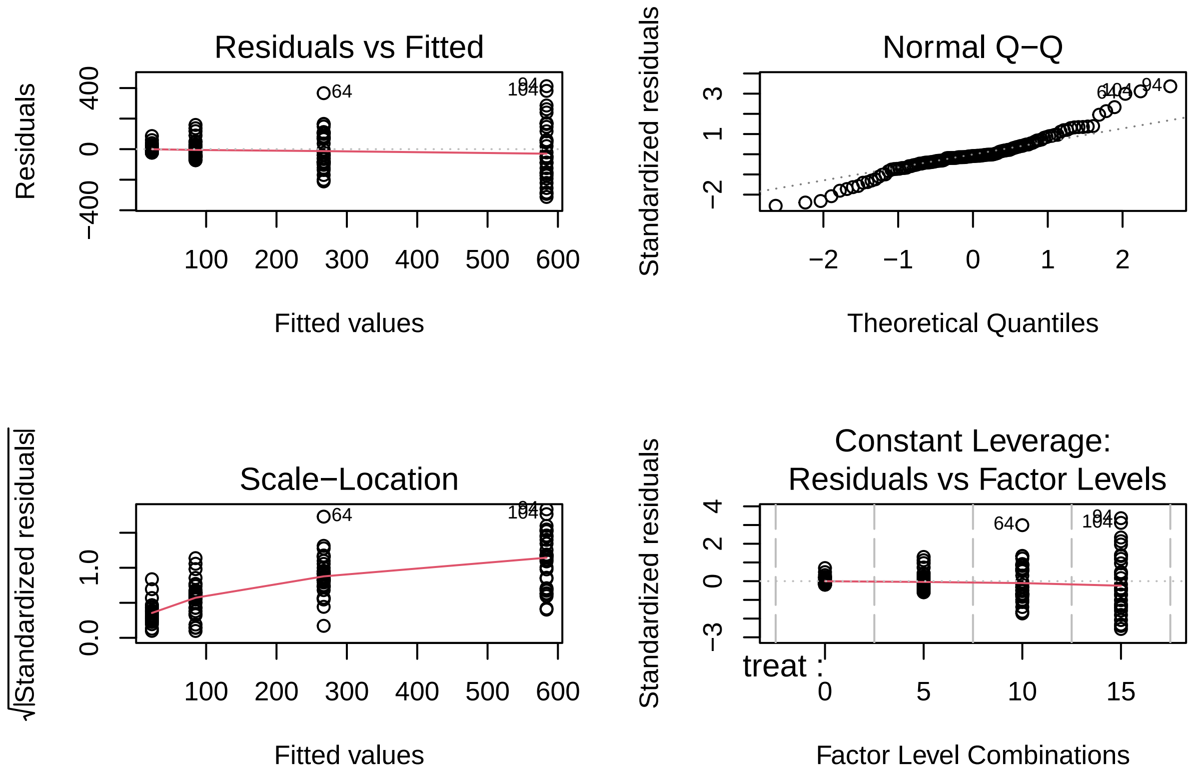
a) What are the ANOVA requirements/assumptions?

- Normality of the residuals

- Independence of the residuals
- Homogeneity of variances
- Linearity
- Additivity (when using an RCBD without replication inside blocks)

b) Based on the diagnostic plots below, are safe using ANOVA? If not, what is the problem and what could be a solution?

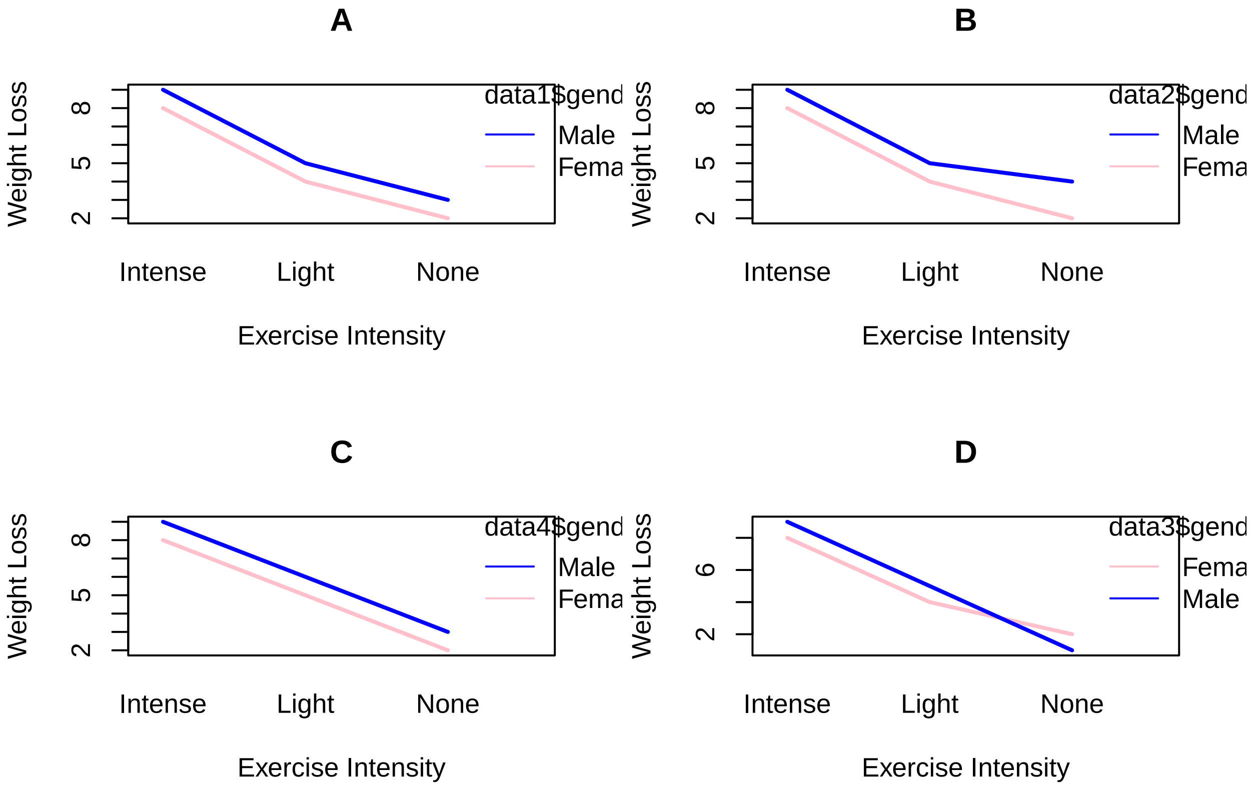
Based on the diagnostic plots below, the residuals don't seem to be homogeneous. We could try to transform the data, to use weighted least squares, or a model that accounts for different variances (such as linear mixed models).



Residuals vs Fitted
Residuals
Fitted values
Normal Q−Q
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Constant Leverage:
Residuals vs Factor Levels
Standardized residuals
treat :
Factor Level Combinations

8. Can we say that there is a significant interaction solely based on the interaction plot? Which of the plots below indicates a possible interaction?

We cannot say the interaction is significant solely based on the interaction plot. We could use the interaction effect in ANOVA (when we have replication), or we could use the additivity test (Tukey's 1 DF test) when we only have one replication. However, in the figures below, plot B and D indicate a possible interaction.



9. Write the statistical model for (explain what each term is):

a) Factorial CRD
$y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + e_{ijk}$
y_{ijk}: Observation of the kth replication, of the ith level of α and the jth level of β.
μ: intercept or grand mean.
α_i: Effect of the ith level of factor α.
β_j: Effect of the jth level of factor β.
$\alpha\beta_{ij}$: Interaction effect of the ith level of α with the jth level of β.
e_{ijk}: Residual associated with the experimental unit ijk.

b) RCBD

$y_{ij} = \mu + \alpha_i + b_j + e_{ij}$
y_{ij}: Observation of the experimental unit ij
μ: intercept or grand mean
α_i: Effect of the ith level of factor α
b_j: Effect of the jth block
e_{ij}: Residual associated with the experimental unit ij

c) CRD

$y_{ij} = \mu + \alpha_i + e_{ij}$
y_{ij}: Observation of the ith level of α in the jth replication

μ: intercept or grand mean
α_i: Effect of the ith level of factor α
e_{ij}: Residual associated with the experimental unit ij

d) Factorial RCBD

$y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + b_k + e_{ijk}$
y_{ijk}: Observation of the kth observation, of the ith level of α and the jth level of β
μ: intercept or grand mean
α_i: Effect of the ith level of factor α
β_j: Effect of the jth level of factor β
$\alpha\beta_{ij}$: Interaction effect of the ith level of α with the jth level of β
b_k: Effect of the kth block
e_{ijk}: Residual associated with the experimental unit ijk

10. Answer the following questions:

a) If the treatment effect is zero, what would be the F value for treatment?

If the treatment effect is zero, the calculated F value will be close to 1. Since the numerator and the denominator will be measuring the same thing (the residual).

b) Testing cultivar effect (3 cultivars) in an anova. What are the null and alternative hypothesis?

$H0 : \mu_1 = \mu_2 = \mu_3$
Ha : the mean of at least one of the 3 cultivars is different than the others.

c) In an anova where block is significant, but I did not account for block. The F value for treatment would be larger or smaller than expected? why?

If there block was not accounted for, the effect of block will inflate the residuals, biasing the result. The F test will be smaller since the denominator of the F test (the residual) will be larger.